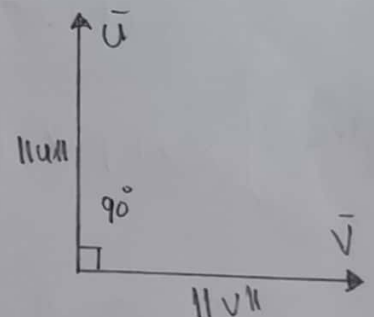


Orthogonality $\Rightarrow$ It is the Property that allows transmission of more than one signal over a common channel with successful detection.

Orthogonal signals : Two signals are said to be orthogonal if they are mutually independent.
فئة ن بمرور متباعدة

Orthogonal Vectors $\Rightarrow$ the two vectors $\vec{u}, \vec{v}$ are orthogonal if and only if inner product between them equal to zero.

$$|\vec{u} \cdot \vec{v}| = \|\vec{u}\| \|\vec{v}\| \cos \theta \quad \Rightarrow \text{when } \theta = 90^\circ \text{ then } \vec{u} \cdot \vec{v} = 0$$



$\rightarrow$ Inner Product acts as Definite Integral
In signal space.

$\rightarrow x_1(t)$ and $x_2(t)$ are orthogonal signals if and only if

$$\int_{-\infty}^{\infty} x_1(t) \cdot x_2(t) \cdot dt = 0$$

for Non-Periodic signal

$$\int_0^T x_1(t) \cdot x_2(t) \cdot dt = 0$$

for Periodic signal.

Properties of Orthogonal signals :-

(2)

- ① Two harmonics of different frequency are always orthogonal i.e.,

$$X_1(t) = \sin(n\omega t + \phi_1)$$

$$X_2(t) = \sin(m\omega t + \phi_2) \quad \text{when } n \neq m$$

$$\int_0^T X_1(t) \cdot X_2(t) \cdot dt = 0$$

- ② Sine and Cosine functions with same phase ϕ and frequency f are orthogonal i.e.,

$$X_1(t) = \sin(n\omega t + \phi)$$

$$X_2(t) = \cos(n\omega t + \phi)$$

$$\int_0^T X_1(t) \cdot X_2(t) \cdot dt = 0$$

- ③ Sine signal and DC signal are orthogonal i.e.,

$$X_1(t) = \text{dc}$$

$$X_2(t) = \sin(n\omega t + \phi)$$

$$\int_0^T X_1(t) \cdot X_2(t) \cdot dt = 0$$

④ $X(t) = X_e(t) + X_o(t)$

$X_e(t)$: even components.

$$\int_{-\infty}^{\infty} X_e(t) \cdot X_o(t) \cdot dt = 0$$

$X_o(t)$: odd components.

So $X_e(t)$ and $X_o(t)$ are orthogonal.

⑤ $\rightarrow$ Average Power for $y(t)$ = Average Power for $x_1(t)$ plus Average Power for $x_2(t)$ if $x_1(t)$ and $x_2(t)$ are orthogonal and $y(t) = x_1(t) + x_2(t)$. (3)

$\rightarrow$ Total Energy for $y(t)$ = total Energy for $x_1(t)$ plus total Energy for $x_2(t)$ if $x_1(t)$ and $x_2(t)$ are orthogonal and $y(t) = x_1(t) + x_2(t)$.

E.x Compute the Average Power for the following signal so

① $y(t) = 2 \sin(3\omega_0 t + 45^\circ) + 4 \sin(4\omega_0 t + 35^\circ)$

② $y(t) = 2 \cos(2\omega_0 t + 45^\circ) + 3 \sin(2\omega_0 t + 45^\circ)$

③ $y(t) = 2 + 4 \sin(3\omega_0 t + 35^\circ)$.

Soln =

① $P_y = P_{x_1} + P_{x_2}$ because $x_1(t), x_2(t)$ are orthogonal

$$= \frac{2^2}{2} + \frac{4^2}{2} \Rightarrow P_y = \frac{4}{2} + \frac{16}{2}$$

$$= 2 + 8 = 10 \text{ watt}$$

② $P_y = P_{x_1} + P_{x_2}$ because $x_1(t), x_2(t)$ are orthogonal

$$= \frac{2^2}{2} + \frac{3^2}{2} \Rightarrow \frac{4}{2} + \frac{9}{2}$$

$$\Rightarrow 2 + 4.5 \Rightarrow 6.5 \text{ watt}$$

③ $P_y = P_{x_1} + P_{x_2}$ because $x_1(t)$ and $x_2(t)$ are orthogonal.

$$= (2)^2 + \frac{4^2}{2} \Rightarrow 4 + \frac{16}{2}$$

$$= 4 + 8 \Rightarrow P_y = 12 \text{ watt}$$

→ Compute the Average Power for $y(t) = x_1(t) + x_2(t)$ when $x_1(t)$ and $x_2(t)$ are sinusoidal but not orthogonal, let's

$$y(t) = A_1 \sin(\omega t + \phi_1) + A_2 \sin(\omega t + \phi_2) \quad (4)$$

$$\text{Average Power for } y(t) = \frac{1}{2} (A_1^2 + A_2^2 + 2A_1A_2 \cos(\phi_1 - \phi_2)).$$

E.X Compute the Average Power for

$$y(t) = 2 \sin(3t) + 3 \sin(3t + 5\pi/6)$$

Sol = Average Power $y(t) = \frac{1}{2} \left[2^2 + 3^2 + 2 \times 2 \times 3 \cos(0 - 5\pi/6) \right]$

$$= \frac{1}{2} [4 + 9 + 12 \cos(150^\circ)] = \frac{1}{2} [13 - 10.392]$$

$$= 1.303 \text{ watt}$$

Q. Prove that total power equal to $\frac{1}{2} [A_1^2 + A_2^2 + 2A_1A_2 \cos(\phi_1 - \phi_2)]$ for $A_1 \sin(x + \phi_1) + A_2 \sin(x + \phi_2) = A_0 \sin(x + \phi)$?

Sol = In General Average Power = $\frac{A_0^2}{2} = \frac{1}{2} [A_1^2 + A_2^2 + 2A_1A_2 \cos(\phi_1 - \phi_2)]$

i.e. $A_0 = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\phi_1 - \phi_2)}$

start from:

$$A_1 \sin(x + \phi_1) + A_2 \sin(x + \phi_2) = A_0 \sin(x + \phi) \quad \text{--- (1)}$$

we know that =

$$\sin(x + \phi_1) = \sin x \cos \phi_1 + \cos x \sin \phi_1$$

$$\sin(x + \phi_2) = \sin x \cos \phi_2 + \cos x \sin \phi_2$$

$$\sin(x + \phi) = \sin x \cos \phi + \cos x \sin \phi$$

the equation (1) becomey :-

$$A_1 [\sin \alpha \cos \phi_1 + \cos \alpha \sin \phi_1] + A_2 [\sin \alpha \cos \phi_2 + \cos \alpha \sin \phi_2] \\ = A_0 [\sin \alpha \cos \phi + \cos \alpha \sin \phi]$$

after simplify the equation :-

(5)

$$\sin \alpha [A_1 \cos \phi_1 + A_2 \cos \phi_2] + \cos \alpha [A_1 \sin \phi_1 + A_2 \sin \phi_2] \\ = \sin \alpha [A_0 \cos \phi] + \cos \alpha [A_0 \sin \phi]$$

From above equation we get :-

$$A_1 \cos \phi_1 + A_2 \cos \phi_2 = A_0 \cos \phi \text{ --- (2)}$$

$$A_1 \sin \phi_1 + A_2 \sin \phi_2 = A_0 \sin \phi \text{ --- (3)}$$

Squaring the equation (2) and (3)

$$A_1^2 \cos^2 \phi_1 + A_2^2 \cos^2 \phi_2 + 2A_1 A_2 \cos \phi_1 \cos \phi_2 = A_0^2 \cos^2 \phi \text{ --- (4)}$$

$$A_1^2 \sin^2 \phi_1 + A_2^2 \sin^2 \phi_2 + 2A_1 A_2 \sin \phi_1 \sin \phi_2 = A_0^2 \sin^2 \phi \text{ --- (5)}$$

from add equation (4) and (5) we get :-

$$A_1^2 + A_2^2 + 2A_1 A_2 [\cos \phi_1 \cos \phi_2 + \sin \phi_1 \sin \phi_2] = A_0^2$$

we know that :-

$$1 = \sin^2 \phi + \cos^2 \phi$$

$$1 = \sin^2 \phi_1 + \cos^2 \phi_1$$

$$1 = \sin^2 \phi_2 + \cos^2 \phi_2$$

$$\cos (\phi_1 - \phi_2) = \cos \phi_1 \cos \phi_2 + \sin \phi_1 \sin \phi_2$$

So :-

$$A_0^2 = A_1^2 + A_2^2 + 2A_1 A_2 \cos (\phi_1 - \phi_2)$$

$$A_0 = \sqrt{A_1^2 + A_2^2 + 2A_1 A_2 \cos (\phi_1 - \phi_2)}$$

$$\text{and Average Power} = \frac{1}{2} [A_1^2 + A_2^2 + 2A_1 A_2 \cos (\phi_1 - \phi_2)] .$$